MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAM

Model Question Paper

MG5DSCCHE300 - ORGANIC CHEMISTRY 3

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Part A

Multiple Choice Questions, Answer all Question

Each question carries 1 mark

1. Nitroalkanes containing an α-hydrogen exhibit which type of tautomerism? [K, CO1]
   A. Keto–enol tautomerism B. Nitro–aci-nitro tautomerism
   C. Aldo–keto tautomerism D. Amine–imine tautomerism

2. The complete reduction of phenyl isocyanide (C_6H_5–N≡C) gives which of the following compound? [U, CO1]
   A. Aniline B. Benzylamine
   C. N-Methylaniline D. N,N-Dimethylaniline

3. Which of the following cannot be made by using Williamson's Ether Synthesis [U, CO2]
   A. Methoxybenzene B. Benzyl-*p*-nitrophenyl ether
   C. *tert*-butyl methyl ether D. Di-*tert* butyl ether

4. In pyrrole, the lone pair of electrons on nitrogen is located in which type of orbital? [K, CO3]
   A. sp orbital B. sp^2orbital
   C. sp^3 orbital D. unhybridized p orbital

5. Which pair consists exclusively of pyrimidine bases found in DNA? [U, CO3]
   A. Adenine and guanine B. Cytosine and thymine
   C. Adenine and thymine D. Guanine and cytosine

6. The heating of phenyl methyl ethers with HI produces [U, CO2]
   A. Methanol B. Iodobenzene
   C. Phenol D. Toluene

7. Which of the following represents the correct order of aromatic character among the five-membered heterocycles? [A,CO3]
   A. Pyrrole > Furan > Thiophene B. Thiophene > Pyrrole > Furan
   C. Furan > Pyrrole > Thiophene D. Pyrrole > Thiophene > Furan

8. Which of the following is a fused heterocyclic compound? [K, CO3]
   A. Pyridine B.Pyrimidine
   C. Indole D. Furan

9. When diethylmalonate is hydrolysed followed by heating, the product formed is…. [U, CO4]
   A. Acetic Acid B. Propanoic Acid
   C. Butyric Acid D. Formic Acid

10. The radiationless transition between two electronic states of the same multiplicity is known as…. [U, CO5]

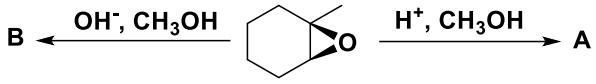
A. Intersystem crossing B. Internal conversion
C. Fluorescence D. Phosphorescence

[1 × 10 = 10]

Part B
Short Answer Type Questions, Answer any four Questions
Each question carries 5 marks

11. a) Explain the mechanism of Hoffmann Bromamide reaction
b) Write one advantage and disadvantage of Gabriel's phthalimide synthesis. [U, CO1]

12. Predict the structure of A and B in the following reaction. Justify your answer. [U, CO2]



13. Suggest one method for the synthesis of Indole. Predict the product of the reaction of indole with the following reagents [K, CO3]
a) Acetyl chloride in presence of $SnCl_4$ b) Reduction with Sn/HCl
b) Reduction with H_2 in presence of Raney Nickel

14. Starting from acetoacetic ester, how will you synthesize [U, CO4]
a) Succinic acid b) Crotonic acid c) Acetic acid

15. Discuss the important differences between photochemical and thermal reactions, with suitable examples. [U, CO5]

16. Explain the molecular events involved in the photochemical process of vision. [U, CO5]

[4 × 5 = 20]

Part C
Long Essay Type Questions, Answer any Two Questions
Each question carries 10 marks

17. a) How is benzene diazonium chloride prepared? Give the mechanism of diazotization.
b) How will you prepare the following
i) o-Bromophenol from nitrobenzene ii) p-Dintro benzene from Aniline [A, CO1]

18. What is Hinsberg's reagent? Explain the Hinsberg's method for the identification of primary, secondary, and tertiary amines. [U, CO1]

19. a) Explain the principle and procedure of Zeisel's method for the estimation of alkoxy groups.
b) Explain briefly on Claisen rearrangement [U, CO2]

20. Outline the mechanism of [U, CO4]
a) Paterno-Buchi reaction
b) Photo-Fries reaction
c) Barton Reaction

[2 × 10 = 20]